

## AVL Trees

Dr. Rabins Porwal

Dr. Rabins Porwal

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## AVL (Adelson-Velskii & Landi) Tree

- Also known as *Height Balanced Tree*
- Def:** Let  $T$  be a non-empty binary search tree with  $T_L$  &  $T_R$  as its left & right subtree.  $T$  is height balanced iff
  - (1)  $T_L$  &  $T_R$  are height balanced
  - (2)  $|h_L - h_R| \leq 1$ , where  $h_L$  &  $h_R$  are heights of  $T_L$  &  $T_R$  respectively.
- Balance Factor:** The balance factor of a node  $n$  in a binary tree is defined to be  $h_L - h_R$ , where  $h_L$  &  $h_R$  respectively, are the heights of left & right subtrees of  $n$ .

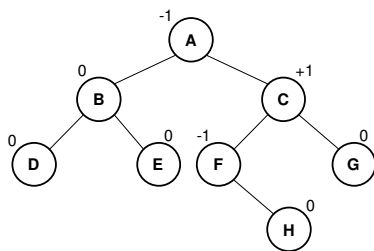
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## Balance Factors: Example



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## Representation:

```

struct AVLNode
{
    int data;
    struct AVLNode * left, * right;
    int balancefactor;
}
  
```

- For any node  $n$  in an AVL tree  $bf(n) = -1, 0$  or  $1$ .
- Note:** The ht. of null tree is defined as  $-1$ .

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## AVL Tree: Advantages

- Since AVL trees are height balanced trees, operations like insertion and deletion have low time complexity.
- Use of AVL trees increase the efficiency of the programs.

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## Insertion in AVL Trees

- While constructing an AVL Tree, a new node can be inserted at any location as per the property of binary search tree and thus tree may become *unbalanced*.
- To make the tree balanced**, some *rotations* are required to perform. These rotations are characterized by the *nearest ancestor, A*, of the inserted node  $y$ , whose balance factor becomes  $\pm 2$ .

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Rotations after insertion in AVL Tree



- The tree is made balanced by performing some rotations:
- Single rotations
  - LL Rotation
  - RR Rotation
- Double Rotations
  - LR Rotation
  - RL Rotation

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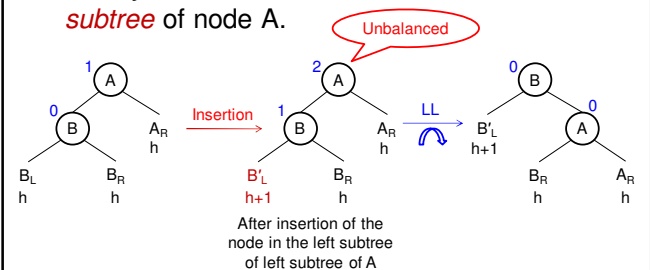
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LL Rotation



- This type of rotation is performed when new node  $y$  is inserted in the **left subtree** of the **left subtree** of node  $A$ .



Unbalanced

After insertion of the node in the left subtree of left subtree of A

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Example



- 50, 40, 30

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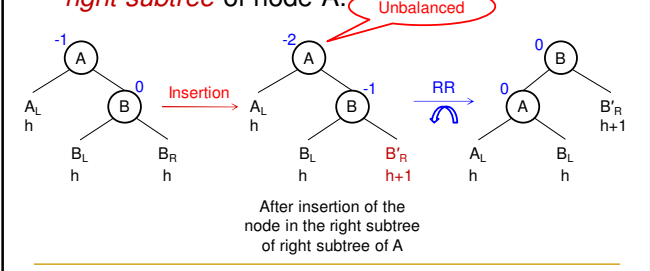
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RR Rotation



- This type of rotation is performed when new node  $y$  is inserted in the **right subtree** of the **right subtree** of node  $A$ .



Unbalanced

After insertion of the node in the right subtree of right subtree of A

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Example



- 50, 60, 70

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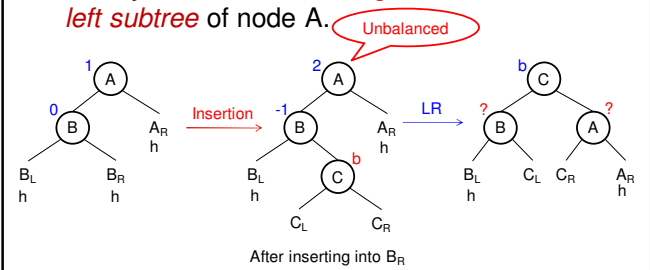
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LR Rotation



- This type of rotation is performed when new node  $y$  is inserted in the **right subtree** of the **left subtree** of node  $A$ .



Unbalanced

After inserting into B<sub>R</sub>

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DATA STRUCTURES

- If  $b = 0$ ,
  - $\Rightarrow \text{bf}(B) = 0 \text{ \& } \text{bf}(A) = 0$ , after rotation
- If  $b = 1$ ,
  - $\Rightarrow \text{bf}(B) = -1 \text{ \& } \text{bf}(A) = 0$ , after rotation
- If  $b = -1$ ,
  - $\Rightarrow \text{bf}(B) = 0 \text{ \& } \text{bf}(A) = 1$ , after rotation

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DATA STRUCTURES

- 50, 40, 45

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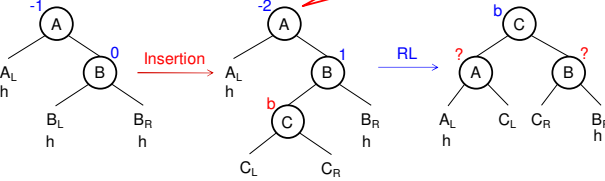


DATA STRUCTURES

### RL Rotation

- This type of rotation is performed when new node  $y$  is inserted in the **left subtree** of the **right subtree** of node  $A$ .

Unbalanced



After inserting into  $B_L$

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DATA STRUCTURES

- If  $b = 0$ ,
  - $\Rightarrow \text{bf}(A) = 0 \text{ \& } \text{bf}(B) = 0$ , after rotation
- If  $b = 1$ ,
  - $\Rightarrow \text{bf}(A) = -1 \text{ \& } \text{bf}(B) = 0$ , after rotation
- If  $b = -1$ ,
  - $\Rightarrow \text{bf}(A) = 0 \text{ \& } \text{bf}(B) = 1$ , after rotation

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DATA STRUCTURES

### Example

- 50, 60, 55

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DATA STRUCTURES

### More Examples

- 50, 40, 35, 58, 48, 42, 60, 30, 33, 25
- 10, 20, 30, 42, 28, 29, 50
- 10, 20, 32, 35, 25, 27

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